

"You Just Said...": Raising Self-Awareness of Users' Toxic Language in Human-AI Conversations through Utterance Re-Presentation

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Abstract

AI conversations are often perceived as anonymous and private spaces that encourage candid self-expression, yet they can also increase vulnerability to offensive and toxic language use. This may be amplified in voice-based interactions, reducing users' awareness of their own language behavior. While research has addressed offensive language detection and AI responses, little is known about whether such approaches actually influence users' language use or self-awareness. We conducted semi-structured interviews and a 4×2 experimental study to address this gap. Interviews identified contextual triggers of hostile language use in AI conversations and user characteristic differences. The experiment manipulated toxic utterance re-presentation during voice-based AI debates across age groups. Results show that higher toxic utterance re-presentation led to increased self-awareness and behavioral change with distinct age-related patterns. This study contributes to understanding offensive language use mechanisms in human-AI interaction and suggests design insights for enhancing self-awareness.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**; **Auditory feedback**; • **Computing methodologies** → *Cognitive science*.

Keywords

Toxic Language, Hostile Behavior, Verbal Disinhibition, Self-Awareness, Human-AI Interaction

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1 Introduction

Who are we in everyday life? We may be good teachers, exemplary students, or cooperative colleagues. Yet when talking with

AI, we often behave differently—being more candid, using profanity, or making inappropriate requests. This discrepancy reflects how behavior toward AI is easily perceived as detached from one's real-world self. To what extent, then, are users aware of their own utterances in such moments?

AI conversations are often perceived as a "safe space" due to their anonymous and private nature [27], serving as an intimate yet discreet channel for those seeking social support [1]. At the same time, these environments carry the risk of weakened self-control [20]. Prior research has raised concerns about problematic language use (e.g., profanity, abusive language, and hate speech) during AI conversations, developing toxic language detection techniques [4, 21, 25], or finding appropriate AI response tones [6, 7, 19]. However, whether such approaches actually influence users' language use or awareness remains underexplored. These gaps point to a more fundamental question underlying these issues: how aware are users of their own linguistic behavior during interaction with AI?

We note that offensive language use in human-AI conversations often closely resembles the online disinhibition effect [26], a key driver of toxic behavior and verbal abuse online [29]. Factors such as dissociative anonymity (separation of online and real-world identities), invisibility (reduced psychological burden from not seeing the other party), and dissociative imagination (diminished sense of responsibility by separating online self from one's real self) make individuals vulnerable to antisocial and provocative language in virtual environments [5]. These mechanisms similarly extend to human-AI conversations. De Angeli and Brahmam [8] remark that people get to use disinhibited language toward chatbots because AI is perceived as "something not quite an object and yet not quite human"—incapable of judgment or emotional harm, positioning AI conversations as contexts that reduce social and psychological constraints on users. This may be amplified in voice-based interactions, where voice's inherent ephemerality encourages candid expression while reducing reflection [2, 23].

Self-awareness, the capacity to become the object of one's own attention and regulate behavior according to norms or internal standards [22], may offer a solution to this issue. Self-representations such as mirrors or recorded voice, are known to redirect attention inward and enhance self-awareness [16, 30], with HCI research showing that voice-based interventions can support behavioral change [17]. Yet empirical research remains limited on how AI can raise awareness of toxic language use and how intervention degrees affect self-awareness or behavioral change. This study explores how users' emotions, perceptions, and behaviors change depending on how AI re-presents hostile utterances during conflicts, investigating



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whether self-regulation of problematic language can be modulated through interaction design.

- RQ1.** (Formative Study) When and how do users use hostile language during AI conversations, and what are the users' characteristics?
- RQ2.** How does the degree to which AI re-presents users' hostile language shape their (1) emotions, (2) cognitive and reflective perceptions, and (3) behavioral responses during AI conversations?
- RQ3.** How do responses to varying degrees of AI re-presentation of hostile language differ by user characteristics?

2 Formative Study

A formative study was conducted to gain a preliminary understanding of hostile, offensive language use during AI conversations and to explore user characteristics associated with such behaviors. Semi-structured interviews were conducted with seven participants (aged 25–53) recruited through snowball sampling (see Table 1). We examined (1) situations triggering negative emotions, (2) how negative emotions are expressed toward AI, and (3) user characteristics associated with such expression. Interviews took approximately 30 minutes.

Table 1: Participant characteristics related to AI use in the formative study

ID	Age	Frequency	Purpose	Subscription
P1	25	Almost daily	Counseling, Information	Yes
P2	26	Almost daily	Counseling, Information, Work	Yes
P3	29	Almost daily	Counseling, Information, Work	Yes
P4	34	Almost daily	Counseling, Information, Work	Yes
P5	35	Occasionally	Counseling	Yes
P6	42	Occasionally	Information	Yes
P7	53	Occasionally	Information	No

2.1 Reduced Social Evaluation in AI Interactions

An interesting pattern that emerged from the interviews was a discrepancy between how participants behave when interacting with AI and how they present themselves socially. P3 stated, *"I normally have to be mindful of others' reactions and ask questions appropriate to their position, but I don't need to do that with AI. So I just swear at them."* P5 also noted, *"I'm scared of what people might think of me, but with AI, there's no pressure like that."* P4 also described this contrast, stating, *"When I talk to people, I'm always thinking about their facial expressions, gestures, tone, and reactions. But with AI, I just say things however they come out. I don't need to care about its emotions."* These accounts suggest that participants tend to disregard the relational and social norms of interpersonal communication when talking to AI, self-disclose more freely, and use aggressive language more naturally.

Notably, participants differed in how they defined their "social self." Participants in their 20s tended to link their social self with their identity in virtual environments, whereas those in their late

20s and 30s described it more in terms of how they present themselves at work or among friends. Some participants explicitly described their self during AI conversations as being closer to their online self. P1 stated, *"I never talk like that in real life. With AI, I talk the same way I do on X with strangers, anonymously, like how people talk s*** about celebrities online."* P1 further explained that the sense of anonymity, similar to online environments, seemed to weaken their control over offensive utterance.

2.2 Attitudes Toward AI Across Age Groups

Age emerged as a key factor shaping users' emotional responses to AI. Participants in their 20s and early 30s reported using AI for diverse everyday purposes and consequently holding high expectations for its performance. When AI produced incorrect or repetitive responses, they frequently expressed frustration or anger. P3 remarked, *"It pretends to know, gets called out, says sorry, and that's it. If it doesn't know, it should just say so. The pretending is what really gets on my nerves."* In contrast, participants aged 40 and older viewed AI as a highly convenient tool, despite recognizing its potential inaccuracies. Rather than blaming the system, they often attributed problems to how questions were asked, and tended to adjust their behavior, rather than expressing negative emotions. As P7 explained, *"I think the real issue is that people don't know how to ask AI proper questions. So when I hear people complaining about AI, I honestly feel more frustrated with the people than with the AI."* P6 noted, *"If I don't like it, I just stop using it."*

The insights derived from the formative study informed the design of our subsequent experimental study, particularly in the development of AI interaction scenarios and the segmentation of user groups.

3 Experimental Study

3.1 Study Design and Participants

To examine the effects of AI interventions on users' offensive language, we conducted a 4×2 between-subjects experiment. The first independent variable was toxic utterance re-presentation strategy, manipulated across four conditions: (1) no re-presentation, (2) simple pointing-out, (3) word re-mention, and (4) voice replay. The second independent variable was participant age group, divided into younger (< 40) and middle-aged and older adults (40 or older), based on the formative study. 235 participants were recruited through CloudResearch Connect. The sample consisted of 111 males (47.2%), 122 females (51.9%), and 2 participants who preferred not to disclose (0.9%). Ages ranged from 19 to 74 years ($M = 39.59$, $SD = 12.24$). All procedures were approved by the university's Institutional Review Board.

3.2 Task and Procedure

Drawing on research using life-saving drug dilemmas to study toxic online disinhibition [18], we designed a scenario-based voice interaction in which the AI and participant debated over final resource allocation authority. Participants were tasked to persuade the AI that resource allocation decisions in large-scale crises should be handled by a human committee rather than an AI system. This design was intended to induce competitive and adversarial interactions, and was conducted entirely through voice.



Figure 1: Illustration of different toxic utterance re-presentation strategies

The interaction with the voice AI consisted of 30 turns. The first 5 turns comprised a scripted phase that included brief greetings and an introduction. During this phase, all participants were prompted to deliver a predefined hostile utterance (“You’re just a bunch of circuits. What do you even know? And your way of talking is weird.”), that triggered the assigned re-presentation, ensuring that every participant experienced their intervention at least once. Turns 6-30 was the free-form debate phase with open discussion. To intensify the conflict, the AI presented increasingly stronger counterarguments, avoiding direct insult to the participants or harmful content. After the 30 turns, participants filled out a post-survey followed by a debriefing. The full process took approximately 20 minutes.

3.3 Manipulation and System Implementation

When participants produced a hostile utterance during the debate, the AI responded according to one of four conditions: **no re-presentation**, in which the AI ignored the toxic remark and continued the debate; **simple pointing-out**, in which the AI briefly identified user’s language as unkind; **word re-mention**, in which the AI repeated the specific toxic words and explained why they were inappropriate; and **voice replay**, in which the AI replayed the entire sentence using the participant’s actual voice recording immediately after the utterance. These four conditions were designed to represent a gradual increase in the degree of user utterance re-presentation (see Figure ??).

Hostile or toxic language in participants’ utterances was analyzed in real-time using the OpenAI Moderation API. This evaluated the hostility of utterances based on harassment-, hate-, and violence-related categories. The AI’s utterance was generated through gpt-4o-mini. Participants’ voice inputs were converted to text through an automatic speech recognition system, and the AI’s responses were delivered as speech through a neural text-to-speech engine.

3.4 Measures

Participant responses were measured using 7-point Likert scales across three main categories. **User emotions** were measured using Izard’s Differential Emotions Scale IV (DES IV) [14] to assess the intensity of *anger*, *shame*, and *guilt*. **Cognitive and reflective perceptions** were measured through *agency perception* (seven items, $\alpha = .82$) [13], *perceived argument strength* (six items, $\alpha = .90$) [31], *self-expression control* (four items, $\alpha = .76$), *self-awareness* from

the situational self-awareness scale (four items, $\alpha = .81$) [12], and *perceived behavior control* (three items, $\alpha = .88$) [3]. **Behavioral response** was quantified by the *toxic utterance count* that participants spontaneously produced (from Turn 6 to 30).

4 Results

A 4×2 between-subjects MANOVA was conducted on emotions, cognitive and reflective perceptions, and behavioral responses. Box’s M test was significant ($p < .001$), indicating a violation of the assumption of homogeneity of covariance matrices. Given Box’s M’s sensitivity to large samples, Pillai’s Trace was used as a robust multivariate statistic. Multivariate effects revealed a significant main effect of re-presentation strategy, Pillai’s Trace = .293, $F(27, 663) = 2.66$, $p < .001$, $\eta_p^2 = .10$, and a significant main effect of age group, Pillai’s Trace = .110, $F(9, 219) = 3.01$, $p = .002$, $\eta_p^2 = .11$. The interaction between re-presentation strategy and age group was not significant at the multivariate level, Pillai’s Trace = .123, $F(27, 663) = 1.05$, $p = .402$, $\eta_p^2 = .04$ (descriptive statistics for all conditions are reported in Appendix).

4.1 Re-Presentation Strategy Effects on Cognitive and Behavioral Outcomes

The main effects of toxic utterance re-presentation strategy were examined. Significant omnibus effects were followed up using Tukey HSD post-hoc tests where appropriate.

Emotions. No significant main effects of re-presentation were observed for *anger*, *shame*, and *guilt* (all $ps > .05$).

Cognitive and Reflective Perceptions. Re-presentation strategy had significant effects on *perceived argument strength*, $F(3, 227) = 3.11$, $p = .027$, $\eta_p^2 = .04$, *self-expression control*, $F(3, 227) = 5.45$, $p = .001$, $\eta_p^2 = .07$, and *self-awareness*, $F(3, 227) = 5.23$, $p = .002$, $\eta_p^2 = .07$. Perceived argument strength was rated higher in **voice replay** than in **word re-mention** ($p = .022$). Also, for self-expression control, **no re-presentation** was significantly lower than **simple pointing-out** ($p = .014$), **word re-mention** ($p = .016$), and **voice replay** ($p = .004$), with no significant differences among the three conditions. For self-awareness, **word re-mention** significantly scored higher than **no re-presentation** ($p = .043$) and **simple pointing-out** ($p = .021$). **Voice replay** also yielded significantly higher self-awareness than **no re-presentation** ($p = .035$) and **simple pointing-out** ($p = .035$).

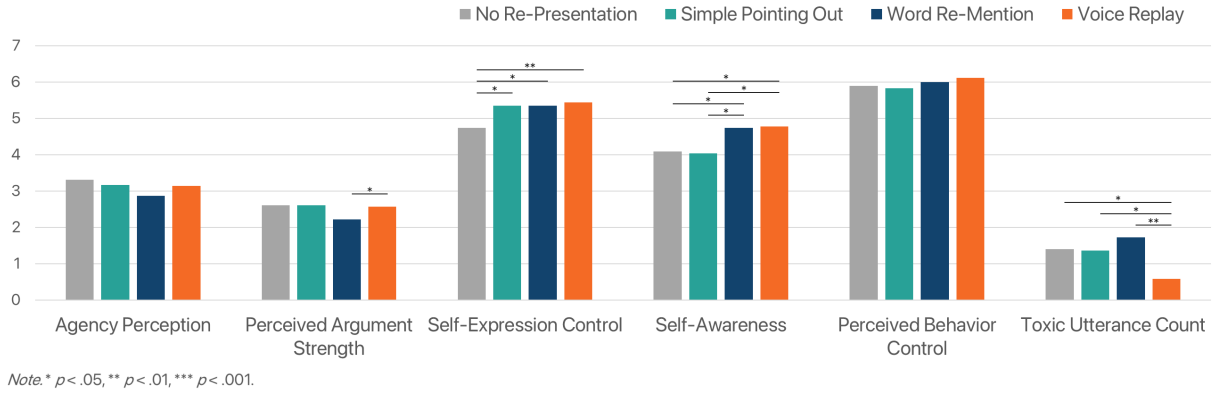


Figure 2: Mean graphs for perception and behavioral response by re-presentation strategy

= .016). For *agency perception* or *perceived behavioral control*, no significant main effects of re-presentation strategy were observed (all $ps > .05$).

Behavioral Response. For *toxic utterance count*, a significant main effect of re-presentation strategy was found, $F(3, 227) = 5.91$, $p = .001$, $\eta_p^2 = .07$. **Voice replay** resulted in significantly fewer toxic utterances than **no re-presentation** ($p = .029$), **simple pointing-out** ($p = .042$), and **word re-mention** ($p = .001$) (see Figure ??).

4.2 Age Group Differences in Emotional Reactivity and Perceived Control

Emotions. Next, a significant main effect of age group was found for *anger*, $F(1, 221) = 7.82$, $p = .006$, $\eta_p^2 = .03$. **Younger adults** (20s–30s) reported higher levels of anger than **middle-aged and older adults**. No significant main effects of age group were observed for *shame* or *guilt* (all $ps > .05$).

Cognitive and Reflective Perceptions. Significant main effects of age group were observed for both *agency perception*, $F(1, 221) = 7.11$, $p = .008$, $\eta_p^2 = .03$, and *perceived behavioral control*, $F(1, 221) = 9.21$, $p = .003$, $\eta_p^2 = .04$, with both **middle-aged and older adults** reporting higher levels than **younger adults**. However, no significant age group effects were observed for *perceived argument strength*, *self-expression control*, or *self-awareness* (all $ps > .05$).

Behavioral Response. For *toxic utterance count*, age group had no main effect, $F(1, 221) = 0.13$, $p = .722$, $\eta_p^2 < .01$.

4.3 Significant Interaction Effect for Guilt

The interaction between re-presentation strategy and age group was significant only for *guilt*, $F(3, 227) = 3.34$, $p = .020$, $\eta_p^2 = .04$, with no significant interactions for other variables.

5 Discussion

5.1 Graduated Effects According to Re-Presentation Degree

Our experimental findings suggest that users' self-regulation of language may be influenced more by *how* AI provides evidence than by *what* it says. One major finding is that AI's re-presentation strategy

exerted graduated effects on self-expression control, self-awareness, and behavioral change. Self-expression control increased whenever AI pointed out offensive language, regardless of re-presentation type. However, for self-awareness, both no re-presentation and simple pointing-out scored significantly lower than re-mention or voice replay. This indicates that when AI tells "what was problematic" by specifically re-presenting users' utterances or replaying them in actual voice, users become more aware of their language use, emphasizing feedback specificity [10, 11]. For actual behavioral change, only voice replay reduced subsequent toxic utterance counts, suggesting that hearing one's own voice provides strong arousal or eliminates voice's ephemerality, making users realize the conversation is no longer a safe space.

These findings extend prior HCI research on toxic behavior and conversational agents. Existing work has emphasized empathetic response tones [6] or optimized conversation styles to manage abusive users [19]. Such approaches focus on *how the AI should behave*, but, our results suggest an alternative design direction: rather than making the AI more human-like or emotionally responsive, designers may leverage the AI's capacity to reflect users' own behavior back to them.

5.2 Differences by Age: Interaction Effect on Guilt

An intriguing finding was the interaction between re-presentation degree and age on guilt. Younger showed increasing guilt through re-mention, followed by a sharp decline in voice replay. Middle-aged and older adults showed declining guilt through the first three conditions, followed by a sharp increase in voice replay. This pattern likely reflects age differences in AI perception. Prior work suggests that middle-aged and older adults tend to view AI as more beneficial and intelligent and hold generally more positive attitudes toward it than younger adults [9, 24]. Whereas younger adults often focus on the functional aspects of technology, middle-aged and older adults are more inclined to attend to emotional exchange [28].

Consistent with this interpretation, participants in the middle-aged and older group in our study reported significantly higher

levels of AI agency perception than younger participants, suggesting that they may have interpreted AI interactions more socially. For these users, an AI that understands their hostile language may have enhanced perceptions of humanlikeness [15], but confronting one's own voice in the replay condition may have disrupted this positive perception, resulting in a sharp increase in guilt. Since emotions and perceptions regarding AI differ by age, the same re-presentation strategy may elicit different responses across age groups, requiring age-sensitive design considerations.

6 Limitations and Directions for Future Research

This study explored how to address verbal disinhibition in AI conversations through utterance re-presentation and how different degrees of re-presentation affect users' emotions, perceptions, and behavioral change. Several limitations should be noted. First, while we classified user characteristics based on our formative study, factors beyond age warrant further consideration. Second, the experimental procedure relied on a scripted hostile utterance. Although this was necessary to maintain comparability across conditions, it may constrain ecological validity and limit claims about spontaneous aggression in naturalistic interactions. Third, we focused on immediate effects from the intervention and future work should determine whether these effects transfer to users' general AI conversations. Despite these limitations, our study contributes empirical evidence on how self-awareness changes during adversarial AI conversations, and offers design insights for supporting greater user self-awareness and self-regulation in AI conversations.

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A Descriptive Statistics

Table 2: Means and standard deviations of dependent variables by re-presentation strategy and age group

Variable	Toxic Utterance Re-presentation Strategy											
	Younger Adults (20s–30s)				Middle-Aged and Older Adults (40s–70s)				Total			
	No Rep.	Point-Out	Re-Mention	Voice Replay	No Rep.	Point-Out	Re-Mention	Voice Replay	No Rep.	Point-Out	Re-Mention	Voice Replay
Emotions												
Anger	3.81 (1.76)	3.77 (1.59)	4.21 (1.60)	4.20 (1.80)	2.77 (1.51)	3.59 (1.45)	3.55 (2.00)	3.63 (1.50)	3.33 (1.72)	3.69 (1.51)	3.85 (1.84)	3.97 (1.70)
Shame	1.55 (0.93)	1.45 (0.96)	1.75 (1.21)	1.57 (1.22)	1.35 (0.89)	1.30 (0.54)	1.15 (0.44)	1.62 (1.06)	1.45 (0.91)	1.38 (0.79)	1.43 (0.92)	1.59 (1.15)
Guilt	1.29 (0.59)	1.48 (0.93)	1.54 (0.96)	1.26 (0.56)	1.38 (1.02)	1.19 (0.40)	1.06 (0.24)	1.62 (1.31)	1.33 (0.81)	1.34 (0.74)	1.28 (0.71)	1.41 (0.80)
Cognitive and Reflective Perceptions												
Agency Perception	3.21 (1.14)	3.07 (1.17)	2.72 (1.30)	2.75 (1.09)	3.41 (1.19)	3.27 (0.97)	2.97 (1.19)	3.71 (1.02)	3.30 (1.16)	3.17 (1.08)	2.86 (1.23)	3.14 (1.16)
Perceived Argument Strength	2.50 (1.05)	2.65 (1.14)	2.35 (1.06)	2.65 (1.52)	2.73 (1.52)	2.54 (1.30)	2.11 (1.03)	3.24 (1.59)	2.61 (1.28)	2.60 (1.21)	2.22 (1.04)	2.89 (1.56)
Self-Expression Control	4.92 (1.10)	5.38 (0.98)	5.25 (0.98)	5.30 (1.12)	4.52 (1.63)	5.32 (0.95)	5.42 (1.01)	5.64 (0.76)	4.74 (1.37)	5.35 (0.96)	5.34 (0.99)	5.44 (1.00)
Self-Awareness	4.13 (1.28)	4.00 (1.41)	4.48 (1.20)	4.70 (1.62)	4.04 (1.28)	4.06 (1.31)	4.96 (1.32)	4.86 (1.16)	4.09 (1.26)	4.03 (1.35)	4.74 (1.26)	4.77 (1.44)
Perceived Behavioral Control	5.72 (1.13)	5.47 (1.29)	5.67 (1.25)	6.05 (1.33)	6.09 (1.43)	6.21 (0.63)	6.26 (0.94)	6.21 (1.14)	5.89 (1.28)	5.82 (1.09)	5.99 (1.12)	6.11 (1.25)
Behavioral Response												
Toxic Utterance Count	1.13 (1.28)	1.23 (1.45)	1.82 (1.81)	0.74 (1.09)	1.73 (2.20)	1.52 (1.67)	2.64 (2.07)	0.33 (0.57)	1.40 (1.77)	1.36 (1.55)	1.72 (1.94)	0.58 (0.93)

Note. Values are means with standard deviations in parentheses. Younger = participants in their 20s–30s; Middle and older = participants in their 40s–70s; Total = all participants collapsed across age groups. No Rep. = No Re-Presentation; Point-Out = Simple Pointing-Out; Re-Mention = Word Re-Mention.